

InsecureBankv2

1. Informasi Umum

- **Nama Peserta** : Aria Fatah
 - **Tanggal Praktikum** : 29 Juni 2025
 - **Nama Praktikum** : Lab Mobile Application Pentest
 - **Target Sistem** : InsecureBankv2
 - **IP/URL Target** :
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2. Tujuan Praktikum

Tujuan dari praktikum ini adalah untuk **mengidentifikasi dan mengeksploitasi berbagai kerentanan umum pada aplikasi mobile**, khususnya pada platform Android. Target dari pengujian ini adalah **InsecureBankv2**, sebuah aplikasi Android

3. Tools dan Bahan

- **Tools Utama:**
 - MobSF (Mobile Security Framework) via Docker
 - Burp Suite (opsional, untuk intercept dan analisis komunikasi aplikasi jika dilakukan dynamic analysis)
-

4. Metodologi Pengujian

Metode pengujian mengacu pada standar **NIST SP 800-115**:

1. **Planning**
 2. **Discovery**
 3. **Attack**
 4. **Reporting**
-

5. Langkah-Langkah Praktikum

<https://github.com/dineshshetty/Android-InsecureBankv2>

1. Aktifkan service MobSF dengan Docker

```
# git clone https://github.com/MobSF/Mobile-Security-Framework-MobSF
# cd Mobile-Security-Framework-MobSF

docker pull opensecurity/mobile-security-framework-mobsf:latest
docker run -it --rm -p 8000:8000 opensecurity/mobile-security-framework-
mobsf:latest
```

[illegible]

2. Download APK InsecureBankv2

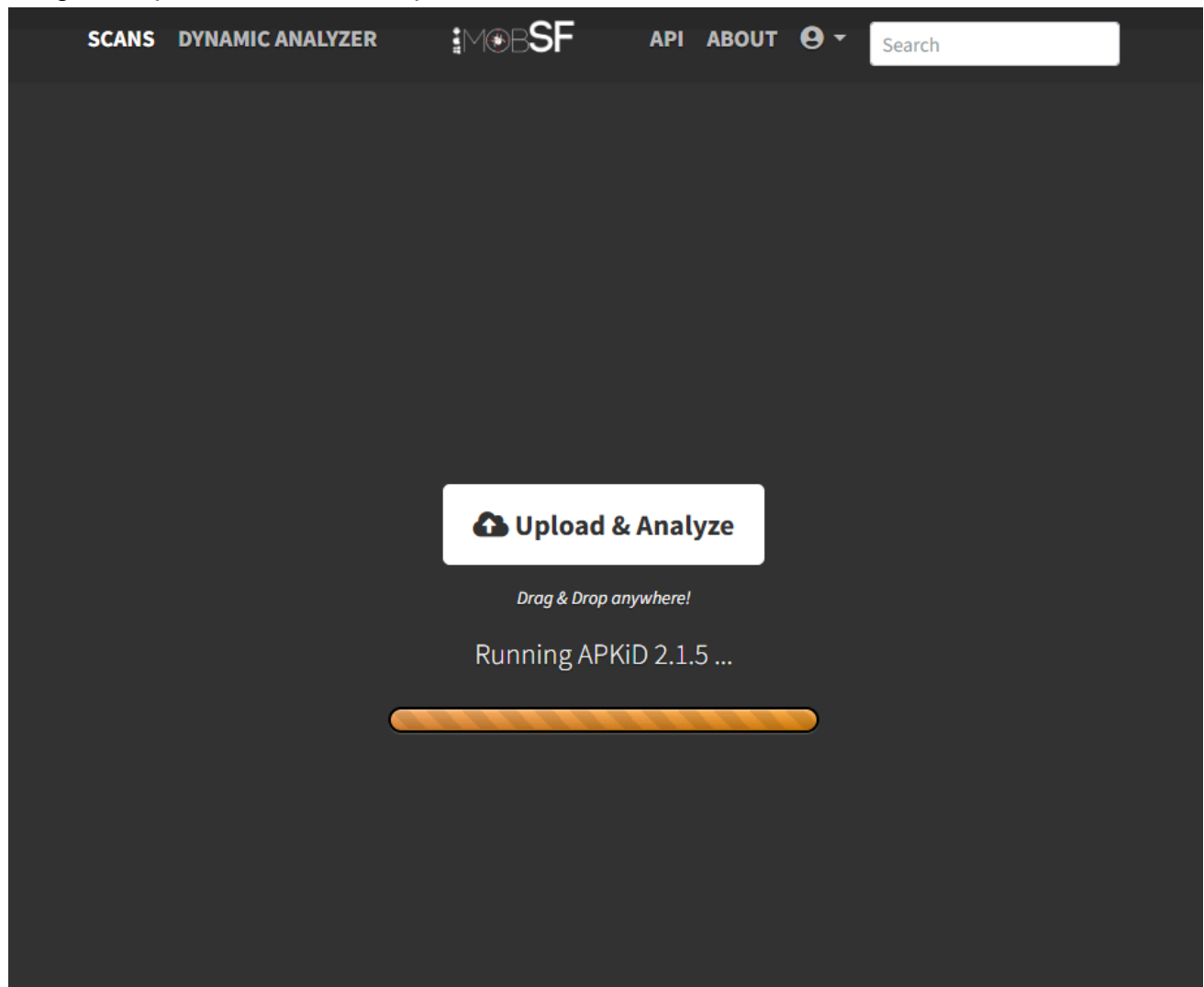
```
git clone https://github.com/dineshshetty/Android-InsecureBankv2
cd Android-InsecureBankv2
# APK ada di folder `InsecureBankv2/bin/InsecureBankv2.apk`
```

Jika tidak ada file APK-nya:

```
# kamu bisa compile sendiri:  
cd Android-InsecureBankv2/InsecureBankv2  
./gradlew assembleDebug
```

3. Analisis APK dengan MobSF

- Buka MobSF di browser: <http://localhost:8000>
- default cred mobsf:mobsf
- Drag & Drop InsecureBankv2.apk ke halaman MobSF



- Tunggu proses static analysis selesai

learn

MobSF

Static Analyzer

Information

Scan Options

Signer Certificate

Permissions

Android API

Browsable Activities

Security Analysis

Malware Analysis

Reconnaissance

Components

PDF Report

Print Report

Start Dynamic Analysis

RECENT SCANS

STATIC ANALYZER

DYNAMIC ANALYZER

API

DONATE

DOCS

ABOUT

Search

APP SCORES

Security Score

33/100

Trackers Detected

3/432

MobSF Scorecard

FILE INFORMATION

File Name

app-release.apk

Size

3.32MB

MD5

fd9eaf5182029014d7c4915e50fb6d2a

SHA1

3166a660c5626c85c3c9a7fd65d89ca94f30df

SHA256

21d91e09112a0195ef52958bdad2b94d0c93c2c8b4d63abf2d64ec7d9590e71

APP INFORMATION

App Name

InsecureBankv2

Package Name

com.android.insecurebankv2

Main Activity

com.android.insecurebankv2.LoginActivity

Target SDK

22

Min SDK

15

Max SDK

Android Version Name

1.0

Android Version Code

1

4 / 10

EXPORTED ACTIVITIES

View All

0 / 0

EXPORTED SERVICES

View All

1 / 2

EXPORTED RECEIVERS

View All

1 / 1

EXPORTED PROVIDERS

View All

SCAN OPTIONS

Rescan

Manage Suppressions

Start Dynamic Analysis

Scan Logs

DECOMPILED CODE

View AndroidManifest.xml

View Source

View Small

Download Java Code

Download Small Code

Download APK

SIGNER CERTIFICATE

Binary is signed
v1 signature: True
v2 signature: True
v3 signature: False
v4 signature: False
X.509 Subject: O=SI, CN=Dns
Signature Algorithm: rsaassa_pkcs1v15
Valid From: 2017-01-25 05:09:33+00:00
Valid To: 2042-01-19 05:09:33+00:00
Issuer: O=SI, CN=Dns
Serial Number: 0h3cc00396
Hash Algorithm: sha256
md5: ab67667f48e4bed7a13d8ceaae26f1d3
sha1: eb54f7d2cb853591ce115ef364930d7cfe9fecaf
sha256: 64ebf1e6428e2d324073dc18b9e9387bac2b467ec335f0dacc67aaa607141be
sha512: 73e8f8f756d9098568746e9c1ee4408464a74ebbd1ad83d02b47f931795408d752cf249cb171bf1e0ef42daaf3b0804418ef3f7e49584ce1e98161e1

4. Result

application permission

APPLICATION PERMISSIONS

Search:

PERMISSION	STATUS	INFO	DESCRIPTION	CODE MAPPINGS
android.permission.ACCESS_COARSE_LOCATION	dangerous	coarse (network-based) location	Access coarse location sources, such as the mobile network database, to determine an approximate phone location, where available. Malicious applications can use this to determine approximately where you are.	
android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all networks.	
android.permission.GET_ACCOUNTS	dangerous	list accounts	Allows access to the list of accounts in the Accounts Service.	
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.	
android.permission.READ_CONTACTS	dangerous	read contact data	Allows an application to read all of the contact (address) data stored on your phone. Malicious applications can use this to send your data to other people.	
android.permission.READ_PROFILE	dangerous	read the user's personal profile data	Allows an application to read the user's personal profile data.	
android.permission.SEND_SMS	dangerous	send SMS messages	Allows application to send SMS messages. Malicious applications may cost you money by sending messages without your confirmation.	
android.permission.USE_CREDENTIALS	dangerous	use the authentication credentials of an account	Allows an application to request authentication tokens.	
android.permission.WRITE_EXTERNAL_STORAGE	dangerous	read/modify/delete external storage contents	Allows an application to write to external storage.	

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CERTIFICATE ANALYSIS

CERTIFICATE ANALYSIS		
HIGH0 WARNING1 INFO1		
Search:		
TITLE	SEVERITY	DESCRIPTION
Application vulnerable to Janus Vulnerability	warning	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.
Signed Application	info	Application is signed with a code signing certificate

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MANIFEST ANALYSIS

MANIFEST ANALYSIS					
HIGH5 WARNING7 INFO0 SUPPRESSED0					
Search:					
NO	ISSUE	SEVERITY	DESCRIPTION	OPTIONS	
1	App can be installed on a vulnerable unpatched Android version Android 4.0.3-4.0.4, [minSdk=15]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.		
2	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.		
3	Activity (com.android.insecurebankv2.PostLogin) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.		
4	Activity (com.android.insecurebankv2.PostLogin) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.		
5	Activity (com.android.insecurebankv2.DoTransfer) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.		
6	Activity (com.android.insecurebankv2.DoTransfer) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.		
7	Activity (com.android.insecurebankv2.ViewStatement) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.		
8	Activity (com.android.insecurebankv2.ViewStatement) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.		
9	Content Provider (com.android.insecurebankv2.TrackUserContentProvider) is not Protected. [android:exported=true]	warning	A Content Provider is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.		
10	Broadcast Receiver (com.android.insecurebankv2.MyBroadCastReceiver) is not Protected. [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.		

MANIFEST ANALYSIS					
HIGH5 WARNING7 INFO0 SUPPRESSED0					
Search:					
NO	ISSUE	SEVERITY	DESCRIPTION	OPTIONS	
11	Activity (com.android.insecurebankv2.ChangePassword) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.		
12	Activity (com.android.insecurebankv2.ChangePassword) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.		

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5. Opsi Analisis Lanjutan

Jika ingin Dynamic Analysis:

- ⚠ Wajib emulator/real device + MobSF Dynamic Analyzer
- Jalankan Android emulator (Genymotion atau AVD bawaan Android Studio)

- Pastikan konek ke MobSF (biasanya MobSF inject device-agent)
- Upload APK dan klik "Dynamic Analysis"